

TrES-3b

R-1801

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210530 · created 2026-08-03T01:57:09+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459364.9388 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.025
Transit depth (Rp/Rs)^2	3.85 +/- 0.98 [%]
Orbital Inclination (inc)	81.32 +/- 0.64
Airmass coefficient 1 (a1)	0.965 +/- 0.011
Airmass coefficient 2 (a2)	0.032 +/- 0.038
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	#3 - [58, 263]
Optimal Method	PSF photometry
Transit Duration (day)	0.0539 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.196 ± 0.025 vs 0.16309 (deviation 0.73σ)
Duration fit vs published	0.0539 d vs 0.0567 d (deviation 0.21σ)
Mid-time O-C	-2.2 min
Depth SNR	4.3
Residual scatter	1.09 %

Light curve

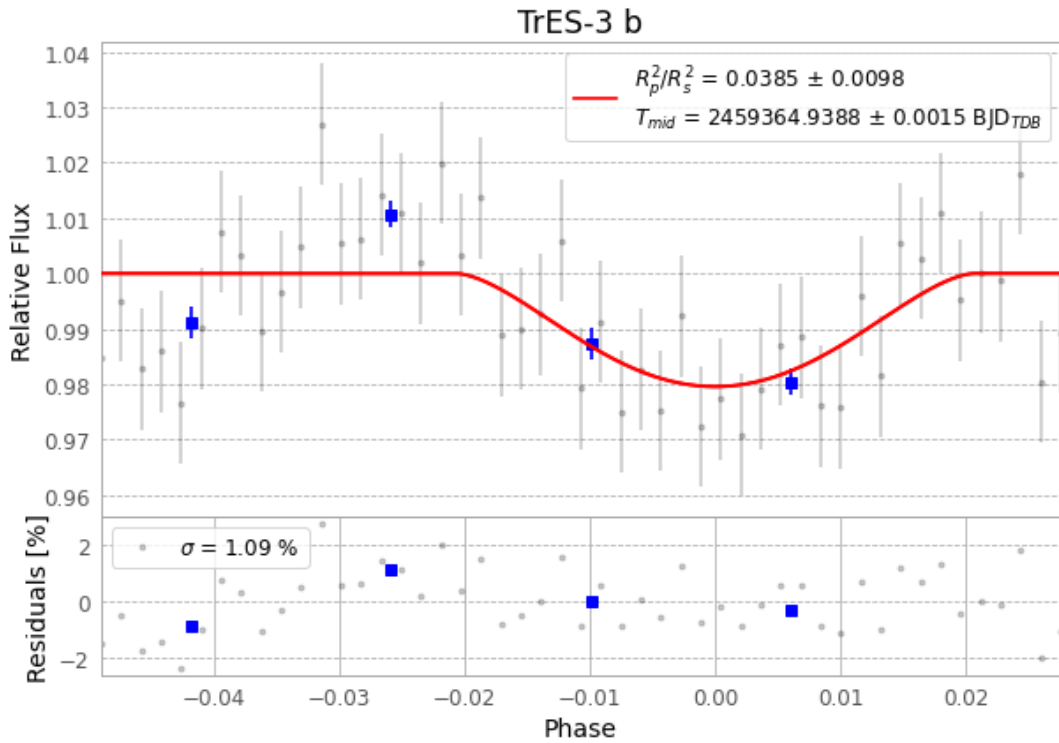


Figure 1 — FinalLightCurve_TrES-3 b_2021-05-30.png (SHA-256 2a6902bb9eeef58...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [398, 222], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 47f9ce0f70d2b6ad...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

49 FITS frames (32 MB), TRES-3210530085412.FITS → TRES-3210530111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210530.log (b0df1969894a...), FinalLightCurve_TrES-3 b_2021-05-30.png (2a6902bb9eee...), FinalLightCurve_TrES-3 b_2021-05-30.csv (df9c01538615...)

Compute resources

Wall time 205.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 01:57Z.